

Seat No. _____

No. of Printed Pages: 03

[82]

SARDAR PATEL UNIVERSITY

External Examination

SYBCA (Semester-IV) (CBCS)

Computer Based Numerical and Statistical Methods (US04FBCA01)

Date: 10/02/2020 (Monday)

Time: 02:00 PM to 05:00 PM

Marks: 70

Q-1 Choose the correct answer from the given options.

[10]

- (1) Let $f(a) < 0$, $f(b) > 0$ and $c \in (a, b)$ is first approximation. If $f(c) < 0$, then in bisection method _____.
- A. a is to be replaced by c B. c is to be replaced by a C. b is to be replaced by c D. c is to be replaced by b
- (2) y depends on x can be written as _____.
- A. $f(x)$ or y_x B. $f(xy)$ C. $f(yx)$ D. None of the these
- (3) The system of linear equation $AX = B$ is said to be non-homogeneous if _____.
A. $B \neq 0$ B. $B = 0$ C. $A \neq 0$ D. A is Symmetric
- (4) The system of linear equation $AX = B$ can be solved by Gauss-Seidel method only if _____.
A. All diagonal elements of A are zero B. All diagonal elements of A are non-zero C. All diagonal elements of A are dominant D. A is skew symmetric
- (5) _____ is not a type of interpolation method.
A. Forward difference B. backward difference C. Newton's divided difference D. moving average method
- (6) Consider the following system of linear equation: $3x + y + z = 0$, $x + 4y - z = 1$, $2x - y + 5z = 2$. If current approximation is $x = 0$, $y = 0.25$, $z = 0.45$, then the Gauss-Seidel method will give next approximation as $x = -0.23$, $y = 0.42$ and $z =$ _____.
A. 0.58 B. 0.48 C. 0.7 D. 0.24
- (7) _____ is called the forward difference operator.
A. Δ B. ∇ C. $\emptyset$ D. U
- (8) Gradual, long-term movement in time-series data is called _____.
A. Seasonal variation B. Cycles C. Trends D. Exponential variation
- (9) If population census for the years 1931, 1941, 1951, 1961 and 1971 is given and if we want to estimate the population for the year 1935 then _____ method is used.
A. Forward difference B. backward difference C. Newton's divided difference D. Lagrangian difference
- (10) Seasonal variation means the variation occurring within _____.
A. a number of years B. parts of a year C. parts of a month D. Part of a week

Q-2 Answer the following questions in brief (Any Ten). [20]

- (1) Define Absolute Error and Relative Error.
- (2) List any two difference between direct and iterative methods.
- (3) Use the False Position method to obtain first approximate solution of the equation $x^3 - 9x + 1 = 0$.
- (4) List out various direct and iterative methods.
- (5) Solve the following system of equations using matrix inversion method.
 $10x + 3y = 5$
 $8x - 2y = 2$
- (6) Explain Interpolation and Extrapolation.
- (7) When will be Newton's backward formula useful?
- (8) Draw the backward difference table.
- (9) Solve the following system of equations using Gauss-Seidel method.
 $20x + 6y = 10$
 $16x - 4y = 4$
- (10) What is time series?
- (11) List the components of time series.
- (12) Differentiate between linear and nonlinear trend.

Q-3 (A) Write an algorithm for Bisection Method. [04]
(B) Find the root of the following equation up to 3 decimal digits using Bisection Method: $X^3 - 4X - 9 = 0$. [06]

OR

Q-3 (A) Explain Newton Raphson Method in brief. [04]
(B) Find the root of the following equation using Secant Method up to 3 decimal places: $X^3 - X - 1 = 0$. [06]

Q-4 (A) Using Lagranges Method of interpolation, obtain $y(2.10)$ [05]

X	2.0	2.25	2.5	2.75	3.0
Y(X)	9.0	10.06	11.25	12.56	14

(B) Using Newton's backward formula, find $y(4.25)$ [05]

X	2.5	3.0	3.5	4.0	4.5
Y(X)	9.750	12.45	15.70	19.25	23.75

OR

Q-4 (A) Using Newton's divided difference formula, find y for $x=4$, [05]

X	-1	0	3	6	7
Y(X)	3	-6	39	822	1611

(B) Using Newton's forward formula, find $y(2.15)$ [05]

X	2.0	2.25	2.5	2.75	3.0
Y(X)	9.0	10.06	11.25	12.56	14

Q-5 (A) Explain the matrix inversion method for solution of system of linear equations in brief. [03]

(B) Solve the following system of equations using Gauss-Seidel method. [07]

$$10X_1 + X_2 + 2X_3 = 44$$

$$2X_1 + 10X_2 + X_3 = 51$$

$$X_1 + 2X_2 + 10X_3 = 61$$

OR

Q-5 (A) The distance covered by a car in given time is given in the following table, determine the speed (dy/dx) of the car at t = 13 minutes. [05]

Time (Minutes)	10	12	14	16	18
Distance (Kilometer)	12	15	20	27	37

(B) Given the following table, find $f'(0.75)$. [05]

x	0.50	0.75	1.00	1.25	1.50
y = f(x)	0.13	0.42	1.00	1.95	2.35

Q-6 Obtain seasonal indices using ratio to moving average. [10]

Year	Q-I	Q-II	Q-III	Q-IV
1970	25	30	21	32
1971	27	28	25	34
1972	22	27	21	30
1973	24	25	20	33

OR

Q-6 (A) Explain any two forecasting models in detail. [04]

(B) Obtain seasonal indices using simple average method. [06]

Year	Q-I	Q-II	Q-III	Q-IV
1974	30	81	62	119
1975	33	104	86	171
1976	42	153	99	221
1977	56	172	129	235

